

Project Report: CLI Task Manager with SQLite - Python Developer

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Family:	Python Developer
Level:	Beginner
Chapters:	6
Source:	CodeQuest project specification (official docs + industry patterns)

Summary

Command-line todo app with SQLite persistence, argparse, and pytest coverage.

Chapters

Track: Project-Intro

Chapter 1: Project Overview [Beginner]

Chapter focus: Project Overview** *(Level: Beginner)

CLI Task Manager with SQLite** is a beginner-level end-to-end project for Python Developer. **Duration:** 1-2 weeks. **Command-line todo app with SQLite persistence, argparse, and pytest coverage.** **Tools:** Python 3.12, sqlite3, pytest.

Code Reference:

```
Objective: CRUD tasks from terminal
Objective: SQLite schema with migrations
Objective: pytest unit tests
Objective: Package with pyproject.toml
```

What it shows: Lists measurable outcomes before you start building.

Topic guide

What it actually is

A project overview defines scope, tools, and success criteria.

When to use it

At project kickoff.

Where to use it

Python Developer portfolios and CodeQuest project submissions.

Real use example

A candidate lists this project on a Python Developer resume with a demo link.

Track: Project-Phase

Chapter 2: Phase: Schema [Beginner]

Chapter focus: Phase: Schema** *(Level: Beginner)

Design tasks table. Complete these tasks: CREATE TABLE; Seed data. Deliverable for this phase: schema.sql.

Code Reference:

```
- [ ] CREATE TABLE
- [ ] Seed data
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Schema is one step in the full project lifecycle.

When to use it

During week 1 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the schema.sql to the teacher.

Chapter 3: Phase: CLI [Beginner]

Chapter focus: Phase: CLI** *(Level: Beginner)

argparse subcommands. Complete these tasks: add/list/done/delete. Deliverable for this phase: Working CLI.

Code Reference:

```
- [ ] add/list/done/delete
```

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase CLI is one step in the full project lifecycle.

When to use it

During week 2 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the Working CLI to the teacher.

Chapter 4: Phase: Tests [Beginner]**Chapter focus: Phase: Tests** *(Level: Beginner)**

pytest for all commands. Complete these tasks: Fixtures; Temp DB. Deliverable for this phase: 90% coverage.

Code Reference:

- [] Fixtures
- [] Temp DB

What it shows: Checklist format helps track phase completion.

Topic guide**What it actually is**

Phase Tests is one step in the full project lifecycle.

When to use it

During week 3 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the 90% coverage to the teacher.

Chapter 5: Phase: Polish [Beginner]**Chapter focus: Phase: Polish** *(Level: Beginner)**

README and packaging. Complete these tasks: pyproject.toml; Entry point. Deliverable for this phase: pip installable package.

Code Reference:

- [] pyproject.toml
- [] Entry point

What it shows: Checklist format helps track phase completion.

Topic guide

What it actually is

Phase Polish is one step in the full project lifecycle.

When to use it

During week 4 of the project timeline.

Where to use it

Python Developer agile sprints and coursework milestones.

Real use example

Team demo shows the pip installable package to the teacher.

Track: Project-Deliverables

Chapter 6: Final Deliverables and Review [Beginner]

Chapter focus: Final Deliverables and Review *(Level: Beginner)**

Submit all final deliverables: GitHub repo; Coverage report; README with usage examples. Skills demonstrated: CLI, SQLite, pytest. Prepare a 5-minute walkthrough video and README for CodeQuest teacher marking.

Code Reference:

```
Deliverable: GitHub repo
Deliverable: Coverage report
Deliverable: README with usage examples
```

What it shows: Final checklist ensures nothing is missing at submission.

Topic guide**What it actually is**

Deliverable review confirms end-to-end project completion.

When to use it

Before deadline and teacher review.

Where to use it

CodeQuest project gallery and interviews.

Real use example

A student submits GitHub repo and passes the rubric.